

BABAYEVO, Russian Federation

WMO: 270080

Lat: **59.4042N**Lon: **35.9114E**Elev: **137**StdP: **99.69**Time Zone: **3.00 (E03)**Period: **90-00**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.9	-24.4	-30.7	0.2	-27.8	-27.1	0.3	-24.2	8.0	-2.7	6.6	-4.0	0.5	340	0.595

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	27.8	18.3	25.7	17.4	23.9	16.4	19.6	24.9	18.5	24.0	17.6	22.4	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	12.9	21.7	16.7	12.1	20.4	15.6	11.2	19.4	56.6	25.0	53.0	23.4	49.9	22.5	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.9	5.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-8.1	-8.0	-3.3	3.5	9.5	15.5	16.6	14.3	8.9	3.5	-4.8	-6.6
	DBStd	10.49	7.10	7.77	4.76	4.84	4.73	4.18	3.00	3.25	4.21	4.68	6.95	6.51
	HDD10.0	2983	561	504	411	201	67	3	1	5	71	206	443	513
	HDD18.3	5459	819	737	669	444	276	107	69	129	284	460	693	772
	CDD10.0	609	0	0	0	7	51	168	206	137	37	4	0	0
	CDD18.3	44	0	0	0	0	3	22	16	3	1	0	0	0
	CDH23.3	503	0	0	0	1	27	244	172	49	9	0	0	0
	CDH26.7	104	0	0	0	0	4	66	32	2	1	0	0	0
Wind	WSAvg	2.1	2.4	2.6	2.5	2.1	2.1	1.9	1.8	1.6	1.7	2.2	2.0	2.4
Precipitation	PrecAvg	644	45	35	34	32	54	75	72	74	59	57	56	51
	PrecMax	766	82	60	67	68	90	129	123	189	130	110	103	84
	PrecMin	433	18	6	7	11	15	12	18	10	30	19	3	23
	PrecStd	83	16	14	17	17	20	30	36	38	25	19	22	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	4.8	9.7	21.5	26.4	30.9	29.9	26.4	24.9	17.4	7.7	3.7
		MCWB	1.9	3.1	5.4	13.8	17.4	18.8	19.5	18.2	17.2	13.5	6.4	3.2
	2%	DB	1.6	3.2	6.4	18.4	23.2	28.6	27.2	24.8	21.0	13.2	5.6	2.6
		MCWB	0.9	2.4	2.6	11.0	14.1	18.7	17.9	16.4	15.6	10.5	4.8	2.1
	5%	DB	1.0	2.0	4.2	15.4	20.8	26.4	25.3	23.1	17.5	11.3	4.4	1.7
		MCWB	0.4	1.2	2.0	9.3	13.3	17.8	17.2	16.2	13.9	9.4	3.8	1.1
	10%	DB	0.5	1.1	2.9	11.8	18.2	23.9	23.4	21.0	15.4	9.7	3.0	0.8
		MCWB	-0.1	0.4	1.3	7.1	12.2	16.8	16.4	15.2	12.6	8.5	2.4	0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	3.6	5.6	15.3	17.7	21.8	21.2	19.5	18.6	13.6	6.7	3.5
		MCDB	2.2	4.6	9.5	21.4	24.8	27.7	25.7	24.1	22.4	16.2	7.4	3.6
	2%	WB	1.1	2.5	3.6	11.8	16.0	19.8	19.4	18.2	16.1	11.3	5.1	2.1
		MCDB	1.4	3.2	5.1	18.0	21.1	25.1	24.9	22.2	19.3	12.7	5.6	2.4
	5%	WB	0.5	1.3	2.5	9.5	14.2	18.7	18.4	17.1	14.4	9.6	3.8	1.1
		MCDB	0.9	1.9	3.7	14.2	18.9	24.7	23.6	21.0	17.2	10.8	4.3	1.5
	10%	WB	0.0	0.4	1.4	7.5	12.8	17.6	17.3	16.0	13.0	8.5	2.5	0.3
		MCDB	0.4	1.0	2.6	11.3	17.5	23.1	21.8	19.9	15.1	9.7	3.0	0.8
Mean Daily Temperature Range	5% DB	MDBR	6.3	7.3	8.3	9.2	10.2	10.5	10.3	9.2	7.8	5.7	5.4	5.4
		MCDBR	4.3	5.0	9.3	15.7	14.8	14.1	13.9	13.6	10.9	8.0	4.4	3.7
		MCWBR	4.2	4.7	6.9	9.5	7.4	6.0	6.3	6.8	6.4	5.7	4.1	3.8
	5% WB	MCDBR	4.7	5.3	6.8	14.3	12.5	12.1	11.9	10.6	10.2	6.7	4.4	4.0
		MCWBR	4.7	5.1	5.3	9.2	7.3	5.9	6.9	6.2	6.5	5.2	4.1	4.1
Clear-Sky Solar Irradiance	taub		0.266	0.307	0.327	0.372	0.363	0.359	0.392	0.385	0.351	0.329	0.294	0.249
	taud		2.076	2.073	2.131	2.228	2.406	2.441	2.359	2.363	2.437	2.391	2.190	2.041
	Ebn at Noon		508	675	793	825	866	875	831	808	779	673	477	384
	Edh at Noon		48	83	106	116	105	103	109	101	80	62	44	34
All-Sky Solar Radiation	RadAvg		0.35	1.03	2.50	3.84	5.06	5.36	5.19	4.07	2.39	1.00	0.31	0.18
	RadStd		0.05	0.23	0.34	0.47	0.38	0.35	0.67	0.32	0.27	0.18	0.05	0.03